MTH 5000, Lecture 10 & 11

LSTM Neural Networks for Time Series Forecasting

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Traditional Forecasting: ARIMA / SARIMA

ARIMA / SARIMA (Statistical Models)

- Assume **linear relationships** in time series data
- Require **stationarity** (constant mean and variance)
- SARIMA extends ARIMA by adding **seasonality**
- Strong for short-term, smooth, and structured patterns

Limitations for Stock Markets

- Struggle with **nonlinear behavior**
- Sensitive to structural breaks (market crashes, shocks)
- Limited ability to model complex dependencies

Machine Learning Forecasting: XGBoost vs Why Not Enough

XGBoost (Machine Learning Approach)

- Converts time series into **supervised learning problem**
- Uses engineered features:
  - Lag values (t-1, t-2, ...)
  - Moving averages
  - Technical indicators (RSI, MACD)
- Captures **nonlinear relationships** better than ARIMA

Limitations for Stock Forecasting

- Does not inherently model **temporal structure**
- Requires heavy feature engineering
- Limited memory of long-term dependencies
- Performance depends strongly on feature design

Key Insight:

- We need models that learn sequences directly

What is a Neural Network?

Neural Networks are computing systems inspired by how the human brain processes information.

Basic Idea:

- Made up of layers of interconnected "neurons"
- Each neuron takes inputs, applies weights, and produces an output
- Learns patterns from data by adjusting weights during training

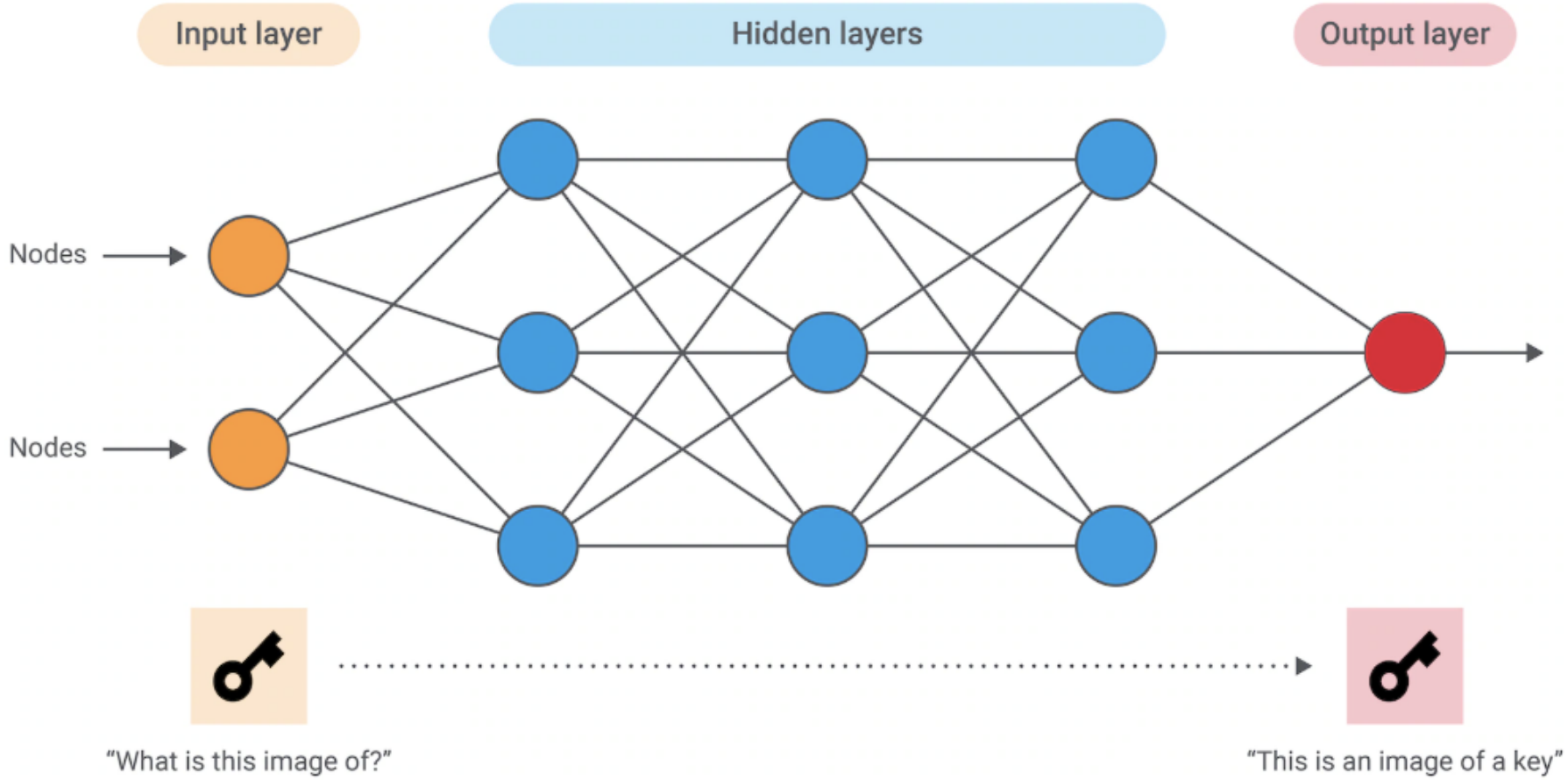
Structure:

- **Input layer:** receives raw data (e.g., stock prices)
- **Hidden layers:** extract patterns and features
- **Output layer:** produces prediction (e.g., future price)

Key Insight:

- Neural networks learn complex nonlinear relationships from data

What is a Neural Network?



What is a Recurrent Neural Network (RNN)?

Recurrent Neural Network (RNN) is a type of neural network designed for **sequential data**.

Key Idea:

- Unlike traditional neural networks, RNNs have a **memory**
- They process inputs one step at a time
- Each step depends on previous computations

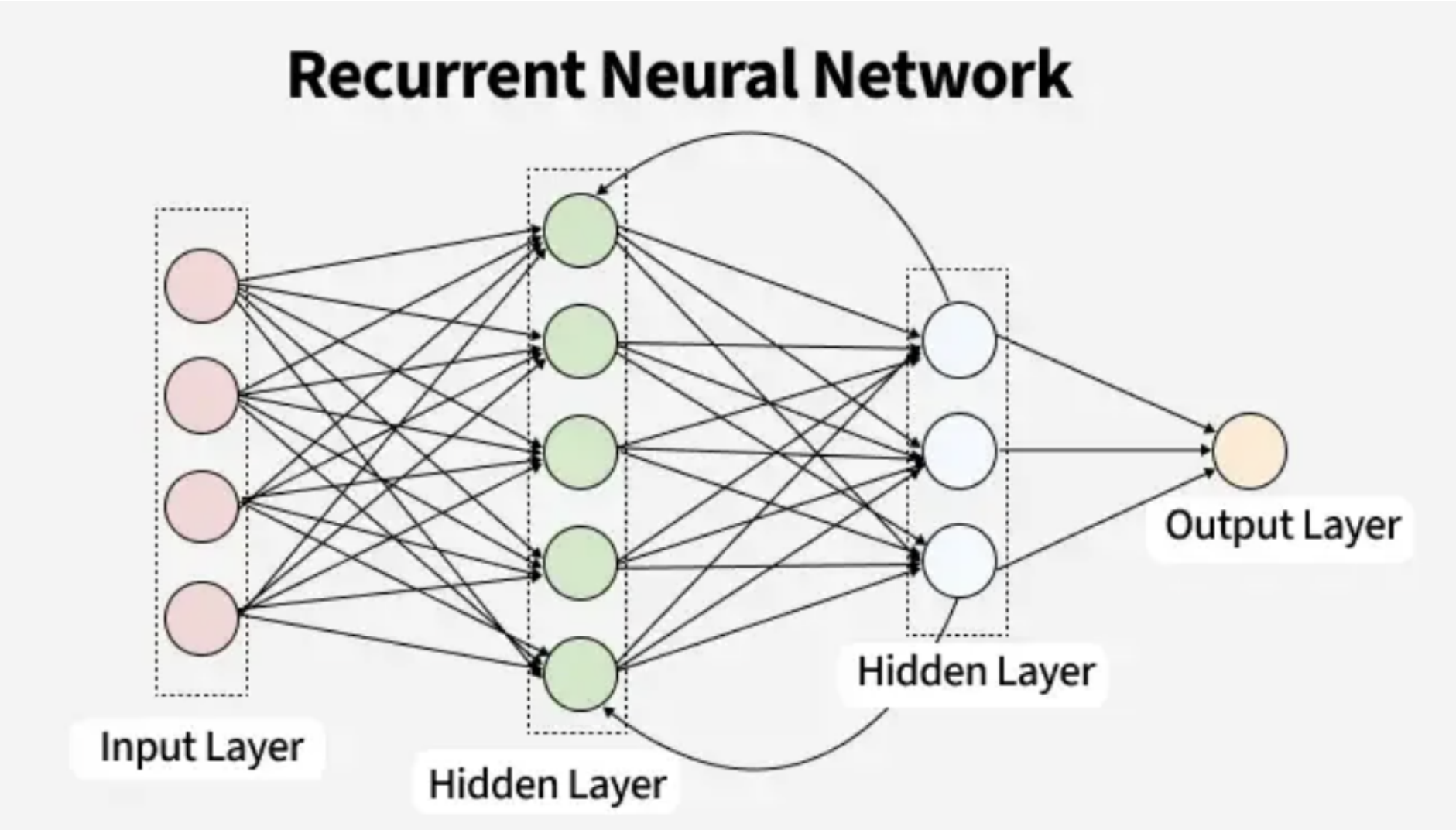
How it works:

- Input at time t: x_t
- Hidden state (memory): h_t
- Output depends on current input + previous hidden state

Why it matters:

- Captures **temporal dependencies**
- Useful for time series, text, and speech data

What is a Recurrent Neural Network (RNN)



What is an LSTM and How Does it Work?

LSTM (Long Short-Term Memory) is a special type of **Recurrent Neural Network (RNN)** designed for sequential data like time series.

Why LSTM?

- Designed to learn **long-term dependencies**
- Solves the **vanishing gradient problem** in standard RNNs

How it works (Intuition):

- Uses a **memory cell** to store important information over time
- Controls information flow using three gates:
  - **Forget gate:** removes irrelevant past information
  - **Input gate:** adds new relevant information
  - **Output gate:** produces the final prediction

Key Idea:

- LSTM acts like a "smart memory system" that learns what to remember and forget

LSTM Architecture (Simplified)

1. **Input Layer (Time Series Window)**
   - Uses past stock data as input sequence (e.g., last 60 days)
   - Each timestep contains features like:
     - Closing price
     - Volume
     - Technical indicators (optional)
   - Data is reshaped into sequences for the model

2. **LSTM Layers (Feature Extraction Over Time)**
   - Processes data step-by-step through time
   - Learns patterns like trends, momentum, and reversals
   - Maintains a hidden "memory" of past information

3. **Dense (Fully Connected) Layer**
   - Takes learned temporal features from LSTM
   - Combines them into a final representation
   - Maps features to prediction space

Stock Price Forecasting Pipeline & Data Preparation

Goal: Transform raw stock data into a format an LSTM can learn from.

1. **Data Collection**
   - Gather historical stock data (Open, High, Low, Close, Volume)
2. **Preprocessing**
   - Normalize values (Min-Max scaling)
3. **Sequence Creation (Sliding Window)**
   - Input: [Day 1, ..., Day 60] $\rightarrow$ Output: Day 61
4. **Tensor Format**
   - Reshape into (samples, timesteps, features)
5. **Training & Prediction**
   - Train LSTM model and predict next value

Model Training

Goal: Train the LSTM to minimize prediction error on future stock prices.

1. **Loss Function (What we optimize)**
   - Mean Squared Error (MSE)
   - Measures average squared difference between predicted and fit price
2. **Optimizer (How learning happens)**
   - Adam optimizer (Adaptive Moment Estimation)
   - Automatically adjusts learning rates during training
   - Efficient for noisy and non-stationary data like stocks
3. **Training Process**
   - Model is trained over multiple passes through data (**epochs**)
   - Typical range: 20–100 epochs depending on dataset size
4. **Batch Training**
   - Data is split into small batches instead of full dataset
   - Improves stability and speeds up convergence and reduce overfitting

LSTM for Stock Forecasting: Advantages & Limitations

Advantages

- Captures sequential patterns in time series data
- Learns long-term dependencies automatically
- Performs well on noisy and non-linear financial data

Limitations

- Cannot guarantee accurate stock market predictions
- Sensitive to hyperparameters (learning rate, layers, etc.)
- Requires large amounts of training data
- Financial markets are highly stochastic and unpredictable

Key Insight:

- LSTM improves pattern learning, but does not eliminate market uncertainty

Key Takeaways

- LSTM is powerful for sequential data
- Stock forecasting is a common use case
- Good preprocessing is critical
- Always validate results carefully